

CoaDTI: multi-modal co-attention based framework for drug–target interaction annotation

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Abstract

Motivation: The identification of drug–target interactions (DTIs) plays a vital role for *in silico* drug discovery, in which the drug is the chemical molecule, and the target is the protein residues in the binding pocket. Manual DTI annotation approaches remain reliable; however, it is notoriously laborious and time-consuming to test each drug–target pair exhaustively. Recently, the rapid growth of labelled DTI data has catalysed interests in high-throughput DTI prediction. Unfortunately, those methods highly rely on the manual features denoted by human, leading to errors. **Results:** Here, we developed an end-to-end deep learning framework called CoaDTI to significantly improve the efficiency and interpretability of drug target annotation. CoaDTI incorporates the Co-attention mechanism to model the interaction information from the drug modality and protein modality. In particular, CoaDTI incorporates transformer to learn the protein representations from raw amino acid sequences, and GraphSage to extract the molecule graph features from SMILES. Furthermore, we proposed to employ the transfer learning strategy to encode protein features by pre-trained transformer to address the issue of scarce labelled data. The experimental results demonstrate that CoaDTI achieves competitive performance on three public datasets compared with state-of-the-art models. In addition, the transfer learning strategy further boosts the performance to an unprecedented level. The extended study reveals that CoaDTI can identify novel DTIs such as reactions between candidate drugs and severe acute respiratory syndrome coronavirus 2-associated proteins. The visualization of co-attention scores can illustrate the interpretability of our model for mechanistic insights. **Availability:** Source code are publicly available at <https://github.com/Layne-Huang/CoaDTI>.

Keywords: Drug–target interaction, co-attention, multi-mode, deep learning

Introduction

Drug–target interaction (DTI) annotation is an essential task for drug development. Unveiling novel protein–ligand binding sites aids in identifying the potential side effect [1], toxicities [2] and drug repositioning [3]. Although conventional manual DTI annotation approaches remain reliable, it is notoriously laborious and time-consuming to test each drug–target pair since therapeutic effects are preferred to be validated by manual laboratory and clinical trials [4]. Moreover, there are more than 97 million candidate compounds for possible target identification in a given drug database [5], demanding massive labour hours.

The state-of-the-art (SOTA) methods including structural approaches and machine learning approaches [6][7] are devised to predict the affinity of DTI automatically and accurately, providing potential DTI candidates for drug development. Structural

approaches utilize molecular docking which considers chemical interaction affinity to predict the DTI [8]. To compute the corresponding affinity, it requires high-quality three-dimensional (3D) information. However, co-crystal structures of protein–ligand pairs are hard to be determined by the researchers, while high-dimensional structure matrices are always sparse, leading to a large amount of tedious and redundancy computations [6].

Recently, the rapid growth of labelled DTI data, especially the drug–target structural data, has raised interests in adopting machine learning methods for DTI prediction [9][10][6]. The DTI prediction task is considered as a binary classification task that integrates the ligand and target features to extract the latent information, indicating their interactions. Previous studies have attempted to leverage high-level features including chemical and biology descriptors to capture the interaction representation [11]

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[12]. Unfortunately, those methods heavily rely on the manual features which contain intrinsic errors, and are not biologically interpretable due to black box issues[13]. To address this drawback, we directly focus on the low-level information, and dispense with auxiliary features. In such a DTI task, only one-dimensional protein sequence is required to feed into the model. The long and short dependency hidden in sequence data can be captured by recurrent neural networks (RNNs) including gate recurrent unit (GRU) [14][15] and long short-term memory (LSTM) [16]. Recently, transformer [17] has emerged as a promising and robust model to learn the semantic information of sequence-like data by utilizing self-attention to learn the global and local sequence dependency. Besides, it also achieves promising performance on other modal data such as images [18]. The image data need to be flattened to fit in the structure of transformer, while protein primary data are naturally sequenced. Therefore, it is natural to employ transformer to capture the features of a protein sequence.

The drug molecule can also be represented as a linear notation which annotates its annotation structure by employing short ASCII strings. Specifically, the length of a molecular sequence is always less than 100 heavy atoms, which means its structural space is relatively small [6]. Since it can be represented as a sequence form, RNN network family is suitable for capturing the underlying features [6]. Analogous to the NLP task that textCNN is utilized to extract the semantic learning of a word sequence, convolutional neural network (CNN) structure has been employed to mine the hidden local patterns of molecular sequence [19]. Moreover, graph neural network (GNN) [20] has been widely applied in drug discovery, including DTI [3], molecular property estimation [21] and synthesis [22]. A graph that consists of edges and nodes is fed into the GNN model. In the DTI prediction problem, the atoms and chemical bonds of a compound are represented as nodes and edges, and can be extracted from the drug molecule sequence. Compared with the aforementioned machine learning models, GNN is more intuitive in representing molecules as it can incorporate the structure information by introducing inductive bias, enhancing the ability of learning the functional relevance of latent space [23]. Traditional GNN only extracts individual node representation. Instead, GraphSage can generate node embeddings by leveraging its neighbourhood information which is suitable for unseen nodes [24]. Novel drug molecules may have different structures that are not included in the training dataset. Incorporating GraphSage can enable the model to recognize unseen molecular structures.

The representation of protein and drug can be regarded as different data modes. Therefore, the model designed for multi-modal tasks has the potential to boost performance. Previous studies have concatenated the representations of protein and drug molecules, and fed the concatenated vectors into fully connected layers [6][3]. However, the single concatenation process cannot estimate which part of the drug compound contributes most to the interaction with the target protein, ignoring cross-modality complementarity. The attention mechanism in deep neural networks has gained great success in both uni-modal tasks and multi-modal tasks since the attention mechanism to multi-modal information fusion is first proposed for visual-question-answering (VQA) tasks. It has been proved that co-attention can learn the fine-grained representation of both image and question [25]. Although researchers have employed a single attention layer to extract the interactions of the learned vectors [3][6], few attempt to adapt deep simultaneous learning and co-attention to DTI problems.

To address those issues, we propose an end-end deep learning framework called CoaDTI for predicting DTI directly based on low-level multi-modal representation and deep co-attention mechanism. The input protein sequence is fed into transformer to obtain the global representation at the amino acid level, while the representation of drug molecules is learned by feeding SMILES (Simplified molecular-input line-entry system) sequences into GraphSage [24]. Inspired by the co-attention mechanism in VQA tasks [25], we improve the self-attention mechanism to model the intra-modal interactions and co-attention to model the inter-modal interactions. Transfer learning is a promising direction to enrich data features from external knowledge [26]. Hence, we replace transformer in protein modal with the pre-trained transformer to circumvent the scarcity of labelled data, and annotate it as CoaDTI-pro for the sake of brevity. We conducted extensive experiments to compare the performances of CoaDTI and CoaDTI-pro on various datasets with realistic data settings. The results of the experiments demonstrate that CoaDTI outperforms other SOTA models. In particular, CoaDTI-pro boosts the performance to an unprecedented level further. Lastly, the visualization of protein and drug cross-attention scores enables researchers to identify the significant residues and atoms in specific protein-drug interactions, providing meaningful clues for downstream pharmaceutical tasks.

Methods

Overview of the CoaDTI framework

The proposed multi-modal co-attention based framework CoaDTI is outlined in Figure 1. Given the input of a drug SMILES sequence, CoaDTI firstly transfers the linear sequence into r -radius sub-graphs [27] which are derived from the vertex's r radius neighbouring vertexes and edges. The subgraph data are fed into the GraphSage model to obtain the embeddings of the compound by leveraging neighbourhood information. Considering the position information of residues is crucial for extracting protein features, CoaDTI employs two embedding layers to obtain the word embeddings and position embeddings with the sine and cosine functions at different frequencies [17]. CoaDTI incorporates the encoder of transformer to capture the protein representation which contains the hidden global information. The vanilla transformer is further replaced with a pre-trained version. After exploiting the sequence and graph representations, CoaDTI translates the information from one modality to another modality, and fuses the multi-modal information to perform a prediction. In order to map the heterogeneous representation into a common space, and estimate the contributions of which part of the drug compound affects the target protein, CoaDTI incorporates deep co-attention mechanisms to handle the multi-modal features. Specifically, CoaDTI utilizes self-attention to encode drug features and protein features, respectively. Then the two feature vectors go through two co-attention layers, namely DPA and PDA, respectively. The DPA layer looks up the effect of drugs on protein, while the PDA layer estimates the protein attention weights on the drug. We also exploited different co-attention mechanisms as elaborated in Section 3.4. The concatenation layer fully connected layers are employed to map the features into the probability of interaction. The methodology details are described below.

Problem formulation

Our model is derived to predict the interaction of the drug and the target, which is a binary classification task. The input is the drug-target pair (G, P) and outputs $y \in \{0, 1\}$, where $y = 1$ indicates

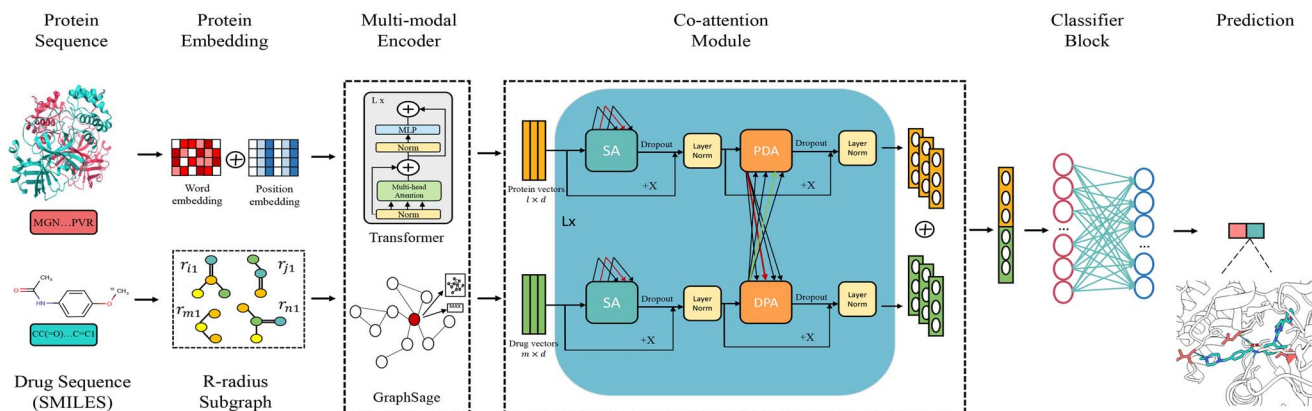


Figure 1. The framework of CoaDTI. CoaDTI has two inputs: protein amino acid sequences and drug SMILES sequences. The amino acid sequence is fed into the word embedding layer and position embedding layer, and the outputs of the two embedding layers are summed together. The summed vector is then inputted into transformer encoder to output the protein feature vector. The SMILES sequence is firstly represented by r-radius subgraphs, and then fed into GraphSage graph to obtain the drug feature vector. CoaDTI incorporates the co-attention module to learn the interaction information. Here, the co-attention module is the intersection version. The detail of the other two modules (stack and encoder-decoder) is described in Section 3.4. The classifier block contains three fully connected layers with Tanh activation function.

there is an interaction between the input drug and input protein. The hidden weights of the co-attention layer are analysed in Section 5.7.

Concretely, the input protein is represented as an amino acid sequence $\mathcal{P} = \{a_1, a_2, \dots, a_n\}$, where a_i is the i^{th} index of the 23 amino acids. Then the protein sequence is fed into the protein embedding layers which map the protein sequence into the latent feature space. The input drug is represented as the SMILES sequence which contains the structure information. Thus, we extract the edges of the atoms by finding the chemical bonds. Then the adjacent matrix is constructed by the edges. The atoms are assigned with the index according to the chemical element type. Finally, we obtain the chemical structure graph $\mathcal{G} = (v, \xi)$, where v represents the atoms, and ξ represents the undirected edges between atoms which are the chemical bonds.

Drug and target representation

SageGraph for drug representation

The drug molecule is represented as a SMILES (Simplified Molecular Input Line Entry System) string. For each SMILES string, CoaDTI extracts the atoms and edges information from the sequence that contains the connectivity and structure information. The underlying information in the sequences is distributed in non-Euclidean space. Inductive biases enable neural networks to impart structure to representations. For this sake, it is intuitive to employ GNN to extract high-level node representations by information propagation. Directly treating the atoms as nodes and chemical bonds as edges is the most explicit way to construct the graph form data. However, molecular graphs consist of a few types of atoms and chemical bonds, leading to fewer learning parameters in the graph model. Hence, r-radius subgraph method [27] is employed to represent the compound graph to tackle the limitation. Supplementary data provide the details.

As a spatial-based convolutional GNN, GraphSage can leverage node features to generate graph embeddings efficiently. Flowing the previous work setting [3], a graph is annotated as $\mathcal{G} = (v, \xi)$. The r-radius subgraph for i vertex is defined as $v_i^{(r)}$. For the sake of simplicity, $v_i^{(r)}$ is replaced by v , and the features of the subgraph are denoted as x_v . We represent the hidden state of the vertex

v at time step t as h_v^t , and the corresponding hidden state of its neighbour subgraph as $h_{\mathcal{N}(v)}^t$. The first time step hidden state h_v^0 is initialized as x_v . The update process is formulated as follows:

$$h_{\mathcal{N}(v)}^t = \max(\{\sigma(\mathbf{W}_{\text{pool}} h_{\mu_i}^{t-1}), \forall \mu_i \in \mathcal{N}(v)\}), \quad (1)$$

$$h_v^t = \sigma(\mathbf{W}_v^t \cdot \text{CONCAT}(h_v^{t-1}, h_{\mathcal{N}(v)}^t)), \quad (2)$$

where $\max$ is the element-wise maximization operator, σ is the nonlinear activation function such as ReLU, $\mathbf{W}_{\text{pool}}$ and $\mathbf{W}_v^t$ are the weight matrices and $t \in \{1 \dots k\}$. The neighbour subgraph information is aggregated in a maximization fashion which depends on the previous time step hidden state information. After obtaining the neighbour information, GraphSage updates the hidden state by concatenating the hidden state at last time step h_v^{t-1} and $h_{\mathcal{N}(v)}^t$ following a fully connected layer with nonlinear activation function σ , which is prepared for the next time step update. The final hidden state h^k is fed into a fully connected layer to get the final drug embeddings $X_d \in \mathcal{R}^{m \times d}$ as the input for the following Co-attention layers, where m is the number of the subgraphs and $d = 512$.

$$X_d = \mathbf{W}_d h^k + b_d, \quad (3)$$

where $\mathbf{W}_d \in \mathcal{R}^d$ is the weight matrix, and $b_d \in \mathcal{R}$ is the bias.

Transformer for protein representation

Proteins are represented as amino acid sequences which are analogous to natural text sequences with limited words. The semantic inferring of natural language processing posits that the words are highly related to the context where they appear. In reality, the order of the amino acids in protein primary structure is determined by DNA, which comprise the polypeptide chain. The covalent peptide bonds are formed by the amine and carboxyl groups of adjacent amino acids. Hence, it is reasonable to speculate that protein primary sequence is also amenable to the hypothesis. In addition to the short distance dependency, considering that proteins fold in 3D space which reflects the long-distance dependency in the linear sequence, we utilize the

encoder of transformer to project the protein sequences to dense vector representations.

As a sequence transduction model, transformer consists of two embedding layers. The first embedding layer is a learned embedding layer that converts the input tokens to vectors of dimension $d_{\text{model}} = 512$. The second embedding layer is the position embedding layer with the same dimension d_{model} . Following the previous work, we adopt sine and cosine functions of different frequencies to calculate the relative position. Then the outputs of the two embedding layers are summed. The encoder of transformer contains multi-head self-attention layers and position-wise feed-forward layers with a residual connection around the two sub-layers [17]. Multi-head attention is described in Section 3.3.3. A layer norm layer is subsequently to apply scalar scale and bias for each entire channel. For each input protein sequence, transformer outputs the protein representation $X_p \in \mathcal{R}^{l \times d}$ (l is the length of protein sequence and $d = 512$).

$$X_{\text{emb}} = X_{\text{learned}} + X_{\text{pos}}, \quad (4)$$

$$X_p = \text{LayerNorm}(x_{\text{emb}} + \text{FFL}(\text{MHA}(x_{\text{emb}}))), \quad (5)$$

where FFL is the feed-forward layer and MHA is the multi-head self-attention layer.

In realistic scenarios, data scarcity undermines the model prediction ability by limiting the model to focusing on a small number of proteins. Growth in the number of protein sequences in public databases has followed an exponential trend over decades, providing vast unlabelled data for self-supervised models which have witnessed great success in language models like BERT [28]. In order to harness the external knowledge, we replace the vanilla transformer with pre-trained transformer, and investigate the performance and robustness improvement. The pre-trained transformer is trained on 86 billion amino acids across 250 million protein sequences spanning evolutionary diversity with 42.6M parameters [29].

Deep co-attention module

Given the dense feature vectors of protein X_p and drug X_d , the next step is to join the information from two modalities to perform the interaction representation. As joint representation methods, previous studies concatenate the individual modality features followed by fully connected layers [6,3]. In this study, we adopt the attention mechanism to integrate information from protein and drug modality with the goal of classification. The idea of the attention mechanism is to use limited attention resources to quickly filter out high-value information from high-volume data. Figure 2 demonstrates three units of attention mechanism in this study, self-attention (SA), drug-protein attention (DPA) and protein-drug attention (PDA). Similar to a natural text sequence, the amino acids in a protein sequence and atoms in a molecular sequence may have long or short dependencies with other amino acids and atoms. Therefore, SA module firstly inputs X_d and X_p into the self-attention mechanism to learn the intra-modal hidden information (Figure 2A). The input of scaled dot-product attention consists of three matrixes, queries, keys and values, which is formulated as follows:

$$Q, K, V = \text{Linear}_Q(X_{\text{emb}}), \text{Linear}_K(X_{\text{emb}}), \text{Linear}_V(X_{\text{emb}}), \quad (6)$$

where $X_{\text{emb}} = X_d$ when feeding drug embeddings and $X_{\text{embedding}} = X_p$ when feeding protein embeddings.

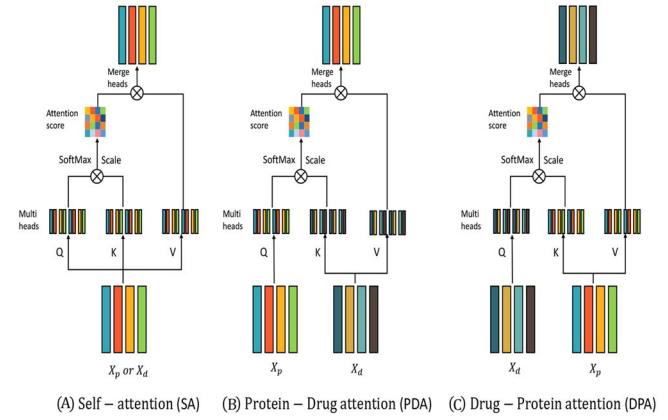


Figure 2. Three versions of the attention layer. (A) Self-attention (SA): the inputs are from one modality. (B) Protein-Drug attention (PDA): the query is the protein feature vector, and the key and value are the drug feature vectors. (C) Drug-Protein attention (DPA): the query is the drug feature vector, and the key and value are the protein feature vectors.

Based on the three matrices, the attention score can be calculated by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad (7)$$

where $\sqrt{d}$ turns the attention matrix into standard normal distribution.

To learn the information from different feature subspaces at different positions, multi-head attention is incorporated into the deep co-attention module. Multi-head attention consists of h heads, and outputs the values. The values are then concatenated and projected again.

$$\text{MHA}(Q, K, V) = \text{CONCAT}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O, \quad (8)$$

$$\text{where } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (8)$$

where $W_i^Q, W_i^K, W_i^V \in \mathcal{R}^{d \times d_h}$ are the projection matrices for the i^{th} head and $W^O \in \mathcal{R}^{hd_h \times d}$. Here, d_h is the dimension of the output from each head, and we employ $d_h = d/h$.

PDA is designed to estimate the contributions of different parts of a drug molecule to the target protein (Figure 2B). Concretely, PDA receives two feature inputs X_p and X_d , where $X_p = [x_{p1}, \dots, x_{pl}] \in \mathcal{R}^{l \times d}$ and $X_d = [x_{d1}, \dots, x_{dm}] \in \mathcal{R}^{m \times d}$. Keys and values are obtained by X_d , and queries are computed by X_p . The multi-head attention learns the pairwise relationship between x_{pi} and x_{di} and outputs the enriched protein vector in accordance with the cross-modal similarity of all the atom features in X_d between x_{pi} by scaled inner dot production. Hence drug embeddings guide the attention learning for protein embeddings in PDA. Analogously, DPA also receives X_p and X_d but measures the effect of protein on drug (Figure 2C). Keys and values are obtained by X_p and queries are computed by X_d .

For all the attention units, the output features X_{update} are fed into a feed-forward layer and dropout layer. The residual connection and layer normalization are incorporated to further improve the robustness of the modules.

$$X = \text{Layernorm}(X + \text{Dropout}(\text{FC}(X_{\text{update}}))). \quad (9)$$

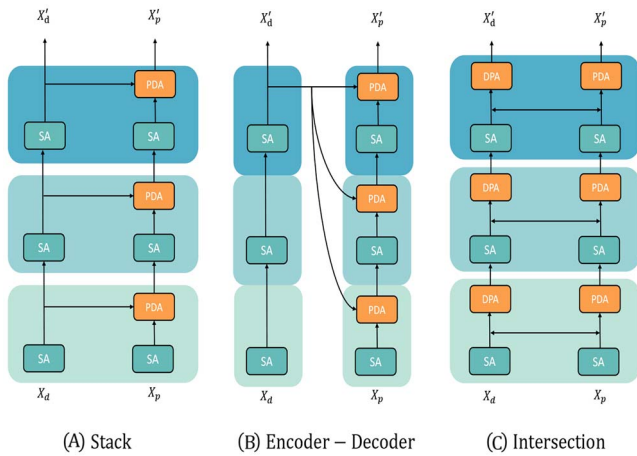


Figure 3. Three versions of co-attention layers. **(A)** Stack layers stack the combination of SA and PDA; **(B)** Encoder-decoder encodes the drug sequence, and decodes the protein sequence. **(C)** Intersection layers stack the combination of SA, PDA and DPA.

Similar to previous work [25], we construct deep co-attention modules by assembling the attention units in different cascade manners (Figure 3). The stack module (Figure 3A) stacks the combination of SA and PDA (SA, SA-PDA). The encoder-decoder module (Figure 3B) borrows the idea of the sequence-sequence model, which is intuitively suitable for the protein and drug sequence feature fusion. The encoder layers learn the drug sequence features by stacking SA layers, and the decoder layers try to translate the information of drug modality to protein modality by stacking SA-PDA layers. The intersection layer (Figure 3C) is devised to exploit the correlation and interactions between drug representations and protein representations. Both the protein and drug input features are inputted by SA layers firstly. Then the protein features are updated by the PDA layer which considers drug effect, while drug features are updated by the DPA layer which considers protein effect. We denote the outputs of the three co-attention modules as $\mathbf{X}_p'$ and $\mathbf{X}_d'$.

Classifier

The output $\mathbf{X}_p'$ and $\mathbf{X}_d'$ are averaged in sequence length dimension to obtain the enriched feature vectors $\hat{\mathbf{X}}_p'$ and $\hat{\mathbf{X}}_d'$. $\hat{\mathbf{X}}_p'$ and $\hat{\mathbf{X}}_d'$ are concatenated to represent the drug-target feature vector. In the classifying block, the interaction vector is fed into the linear layers with the activation function Tanh.

$$f = \text{mean}(\hat{\mathbf{X}}_p') \oplus \text{mean}(\hat{\mathbf{X}}_d'), \quad (10)$$

$$s = \text{Tanh}(W_s f + b_s), \quad (11)$$

where $W_s \in \mathcal{R}^{2d}$ is the weight matrix, and $b_s \in \mathcal{R}$ is the bias.

CoaDTI utilizes the cross-entropy loss as the objective function to minimize.

$$L = -\frac{1}{N} \sum_i y_i \log \sigma(s_i) + (1 - y_i) \log(1 - \sigma(s_i)), \quad (12)$$

where $y_i \in \{0, 1\}$ is the label, which denotes the indicator variable if the relation is the same as the relation of samples, and N is

the number of samples. We minimize the loss by the lookahead optimizer [30] with the inner optimizer SGD.

Experiments

Datasets

We benchmark CoaDTI with different attention modules and CoaDTI-pro with the most suitable attention module (intersection co-attention), and compare the performances with SOTA models on the Human dataset and C.elegans dataset [31]. In addition, we further extend the experiments to BindingDB dataset [32] to investigate the generality and robustness of CoaDTI with real-world data settings.

Human dataset and C.elegans dataset

Human dataset and C.elegans are created by Liu *et al.* [31]. Both the datasets contain highly credible negative samples of compound-protein pairs obtained by a systematic screening framework. Following the assumption that the proteins dissimilar to any known/predicted target of a given compound are not much likely to be targeted by the compound and vice versa, the reliable negative samples are selected by measuring the distance score of the drug and the protein which is calculated by the combined dissimilarities of the proteins and drugs rather than randomly generating DTIs. For a fair comparison, we follow the existing protocol [3] to utilize a balanced dataset, where the ratio of positive and negative samples was 1:1. The balanced Human dataset contains 3369 positive interactions with 1052 unique compounds and 852 unique proteins. The balanced C.elegans dataset contains 4000 positive interactions with 1434 unique compounds and 2504 unique proteins. Following [3], 80% of the data in both two datasets are divided into a training dataset, and the remaining data are considered as development dataset (10%) and test dataset (10%).

BindingDB dataset

BindingDB dataset was created by Gilson *et al.* [32]. BindingDB dataset contains experimentally measured binding affinities with 39 747 positive examples and 31 218 negative examples from a public database. We adopt the dataset constructed by previous work [13] for a fair comparison. The dataset contains 39 747 positive interactions and 31 218 negative interactions. The dataset was pre-divided into train dataset (50 155 interactions), validation dataset (5607 interactions) and test dataset (5508 interactions) by [13].

Performance evaluation metrics

In this study, we adopt the area under the receiver operating characteristic curve (AUC) to evaluate the performance of CoaDTI and CoaDTI-pro. In addition, for Human dataset and C.elegans dataset, we report the values of precision and recall for a fair comparison [3,6]. For BindingDB dataset, we report the AUC and PRC (the area under the precision-recall curve). The detailed metric calculations can be found in Supplementary data.

Results and discussion

Performance on Human dataset and C.elegans dataset

Baseline machine learning models and newly reported deep learning models [3] including K nearest neighbours (KNN), random forest (RF), L2-logistic (L2) and support vector machine (SVM), GCN, GraphDTA [23], CPI-GCN [3] and DrugVQA [6] have been evaluated

Table 1. The results on Human dataset and C.elegans dataset

Methods	Human dataset			C.elegans dataset		
	AUC	Precision	Recall	AUC	Precision	Recall
KNN	0.860	0.927	0.798	0.858	0.801	0.827
RF	0.940	0.897	0.861	0.902	0.821	0.844
L2	0.911	0.913	0.867	0.892	0.890	0.877
SVM	0.910	0.966	0.969	0.894	0.785	0.818
GraphDTA	0.960	0.882	0.912	0.974	0.927	0.912
GCN	0.956	0.862	0.928	0.975	0.921	0.927
CPI-GNN	0.970	0.918	0.923	0.978	0.938	0.929
DrugVQA (VQA-seq)	0.964	0.897	0.948	—	—	—
CoaDTI _{stack}	0.976	0.925	0.951	0.980	0.933	0.941
CoaDTI _{encoder}	0.974	0.925	0.953	0.978	0.951	0.915
CoaDTI _{inter}	0.974	0.922	0.950	0.983	0.944	0.943
CoaDTI-pro	0.982	0.952	0.950	0.985	0.957	0.948

on those datasets. We reported AUC, precision and recall to evaluate the performance of CoaDTI models and other SOTA models on Human dataset and C.elegans dataset. The comprehensive performances of CoaDTI models with different attention modules are also investigated. CoaDTI_{stack} is the model with the stack co-attention module. CoaDTI_{encoder} is the model with the encoder-decoder co-attention module. CoaDTI_{inter} is the model with the intersection co-attention module. CoaDTI-pro utilizes pre-trained transformer to encode the protein features and incorporates the intersection co-attention module. We note that CoaDTI-pro with stack and encoder-decoder modules are not reported since the two models do not show comparative performances with the model equipped with the intersection module. The comparison of CoaDTI-pro with different Co-attention modules is presented in [Supplementary Figure 6](#).

As shown in [Table 1](#), CoaDTI models with different Co-attention modules have close or better performance with the most advanced models on Human dataset and C.elegans dataset. In particular, all the CoaDTI models exceed the SOTA models on AUC. CoaDTI_{stack} achieves the best performance on AUC among the three models, while CoaDTI_{encoder} obtains the highest recall against all the listed models on Human dataset. Furthermore, CoaDTI-pro reaches 0.982, 0.952 and 0.950 on AUC, precision and recall, outperforming the baseline models on all three metrics. Besides, all the CoaDTI models surpass the current models on C.elegans dataset. CoaDTI-pro has the least increases of 0.7%, 2.0% and 2.0% for AUC, precision and recall compared with the current SOTA models. CoaDTI_{inter} shows the best performance among the three models. Although descriptor-based ML models have been proven to averagely achieve better performance than graph-based models on molecular property prediction [33], the low performance of descriptor-based models with AUC (0.860-0.940 and 0.858-0.894) suggests that multi-modal representation extractors in CoaDTI can capture better protein and drug representations for DTI prediction by low-level features. On the other hand, CoaDTI and CoaDTI-pro achieve better performances than the deep learning models; it indicates that co-attention modules enable the model to learn fine-grained protein and drug intra- and inter- information.

Performance on BindingDB dataset

KNN, RF, L2 and SVM are not evaluated on BindingDB dataset since the implementations of those models are not provided in the

Table 2. The results on BindingDB dataset

Methods	AUC	AUPRC
GraphDTA	0.929	0.917
GCN	0.927	0.913
CPI-GNN	0.603	0.543
CoaDTI _{stack}	0.931	0.926
CoaDTI _{encoder}	0.933	0.929
CoaDTI _{inter}	0.941	0.934
CoaDTI-pro	0.959	0.957

reference [3]. The experiment results measured by AUC and PRC are shown in [Table 2](#). Compared with the SOTA models, CoaDTI with different co-attention modules achieve high performance across all metrics. Not only in Human dataset and C.elegans dataset, CoaDTI_{inter} performs the best or competitively among the proposed three CoaDTI models. We speculate that the intersection co-attention layer can measure the interaction on the drug side and the protein side synchronously, and update the cross information iteratively, which can model the real biochemical process to a certain extent. Enhanced by pre-trained protein transformer, the performance of CoaDTI-pro is boosted to an unprecedented level which the AUC and AUPRC are 0.959 and 0.957, respectively.

Novel drug and protein identification

BindingDB dataset is chosen as an extensive dataset to evaluate the generality and robustness of CoaDTI since it focuses on the interactions of proteins and small drug-like molecules. In practice, researchers face the condition that unknown compounds interact with unknown proteins. We have examined CoaDTI and proved that it is able to tackle such a challenging issue. Following the setting in the literature [13], we further split the test dataset into four subsets depending on whether the protein or drug is observed in the training dataset. In the end, we collect *seen drug and seen protein* subset, *seen protein and unseen drug* subset, *unseen protein and seen drug* subset and *unseen protein and unseen drug* subset. We compare CoaDTI-pro with three baseline models including similarity-based method Tiesias [34], deep learning based method DBN [35] consisting of stacked Restricted Boltzmann machines (RBM) and an end-to-end neural network E2E [13] that incorporates LSTM and GCN. We report AUC and accuracy on the four subsets. As

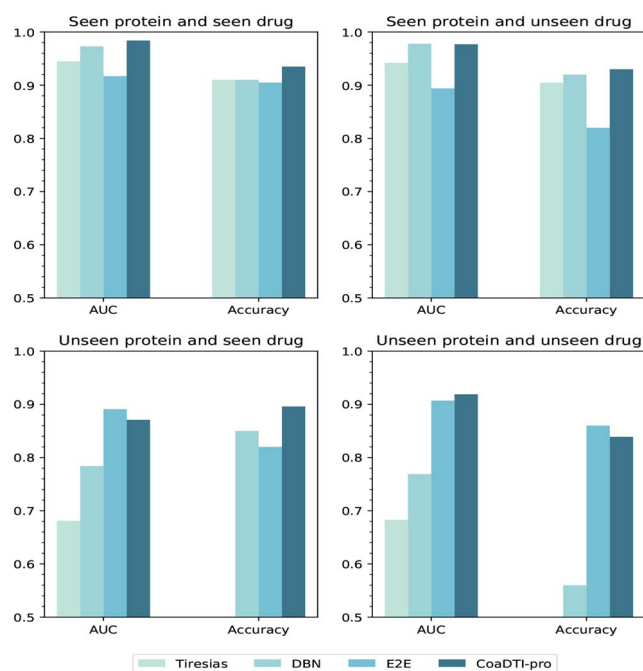


Figure 4. Performance comparison of proposed models, similarity-based approach (Tiresias), stacked restricted Boltzmann layers (DBN) and graph convolutional networks-long short-term memory-based approach (E2E).

depicted in Figure 4, the subsets with unseen protein or unseen drug are relatively difficult to be classified, while CoaDTI-pro consistently shows robust recognition ability across all subsets and all metrics. We provide the description about other models in Supplementary data.

Performance on scarce data

Another real-world issue that should be considered is that the labelled DTI data are limited due to finance and labour restrictions despite the explosion of available unlabelled data. We demonstrated CoaDTI-pro is devised to circumvent this limitation by leveraging the pre-trained transformer which has learned the knowledge hidden in a large amount of unlabelled data which is presented in Figure 5. The details can be found in Supplementary data.

Predicting drug for severe acute respiratory syndrome coronavirus 2 target

To demonstrate the practical implications of CoaDTI, we apply the trained model to annotate the interactions between severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) related proteins and antiviral drugs. In this study, we select SARS-CoV-2 3CL M^{Pro}, SARS-CoV-2 RNA-dependent RNA polymerase (RdRp), SARS-CoV-2 receptor angiotensin-converting enzyme 2 (ACE2), and TMEM16F protein (also known as anoctamin 6) as targets. We employ CoaDTI-pro (trained on Human dataset) to annotate the interactions between SARS-CoV-2-related proteins and candidate drugs. In particular, six candidate drugs are screened for SARS-CoV-2 3CL M^{Pro} and SARS-CoV-2 RNA polymerase, three drugs are screened for ACE2 and 43 drugs for TMEM16F. For a rigorous comparison, Aspirin and Amoxicillin are selected as the negatively controlled drugs since these two drugs are proved to be ineffective against the SARS-CoV-2 [36,37]. In addition to SARS-CoV-2 proteins, we also explore the interactions between candidate drugs and the proteins associated with the host cells

including ACE2 and TMEM16. A detailed description can be found in the Supplementary data.

The aforementioned results suggest that our proposed model can predict interactions between the drugs and SARS-CoV-2 targets accurately. The annotation scores are found for their associations with drug binding affinity. Nonetheless, CoaDTI only relies on the raw amino acid sequence and SMILES, demonstrating the effectiveness and validity of CoaDTI in assisting drug development.

Attention interpretation

Co-attention modules in CoaDTI can enable the model to measure the effect of the drug on the target protein and the effect of protein on the drug in the interaction. Different from the stereotype that a neural network is a black-box model, one merit of our model is that the co-attention module is in line with mechanistic intuition. Instead of time-consuming and costly lab experiments, researchers can gain insights by firstly employing CoaDTI to elucidate where the model has focused on. To perform the interpretability of our model, we select two SARS-CoV-2 3CL M^{Pro} related interactions: GC373 (PDB ID: 6WTK) and ML188 (PDB ID: 7L0D). As we mentioned before, the attention mechanism can generate the attention weights which explain the focus of protein feature vectors and the compound vectors. Figure 6 shows the compound-protein score and protein-compound score of the interaction between GC373 and SARS-CoV-2 3CL M^{Pro}. In particular, the atoms of the compound pay emanative attention to the protein sequence (Figure 6a), while protein vectors focus on the specific atoms of the molecule (Figure 6b). The areas emphasized by the atoms at the top of the SMILES sequence are relatively average, and other atoms are likely to look at the amino acids located left in the protein sequence. The protein sequence focuses on the atoms near the 9th and 27th positions, presenting a pattern of overall attention. The compound-protein score and protein-compound score of the interaction between ML188 and SARS-CoV-2 3CL M^{Pro} are demonstrated in Supplementary Figure 5.

The interactions are depicted in Figure 7 [38][39], where the amino acids with high attention scores are highlighted in red, and the atoms of the compound are highlighted in green. The attention scores for protein are calculated by averaging the protein-compound attention matrix $S_{pc} \in \mathcal{R}^{l \times m}$ on the compound dimension, while the attention scores for the compound are calculated by averaging the compound-protein attention matrix $S_{cp} \in \mathcal{R}^{m \times l}$ on the protein dimension. It can be observed that the protein residues and the molecule atoms that are within or surrounding the binding pocket have high attention values. For the co-crystal complex of 3CL M^{Pro} and GC373 (PDB ID: 6WTK), the key residues identified by the model including HIS41, GLY143, CYS145 and GLU166 overlap substantially with the interaction sites. Two of them (GLY143 and GLU166) form hydrogen bonds with the molecule, and the other residues interact with the molecule by other mechanisms such as van der Waals force and carbon hydrogen bonds. From the drug perspective, hydrogen atoms with sufficient positive charges are the most abundant among the highlighted atoms, which are easier to form hydrogen bonds. In contrast, the hydrogen atoms of the benzene ring are assigned with low attention weights since the hydrocarbon bond of the benzene ring is not a polar bond. The highlighted aldehyde group is active because the electron on the carbon atom is withdrawn by the oxygen atom, and it is also highlighted in red. It can be observed that the aldehyde group forms two hydrogen bonds with the residue CYS145 and the residue GLY143. The carbon atom of the aldehyde group forms the covalent bond with the residue145. An

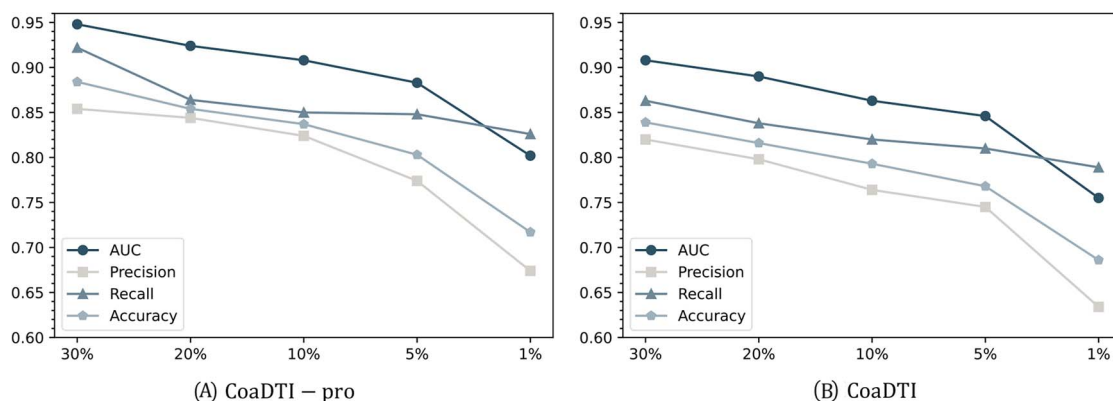


Figure 5. CoaDTI-pro performs better than CoaDTI on scarce data.

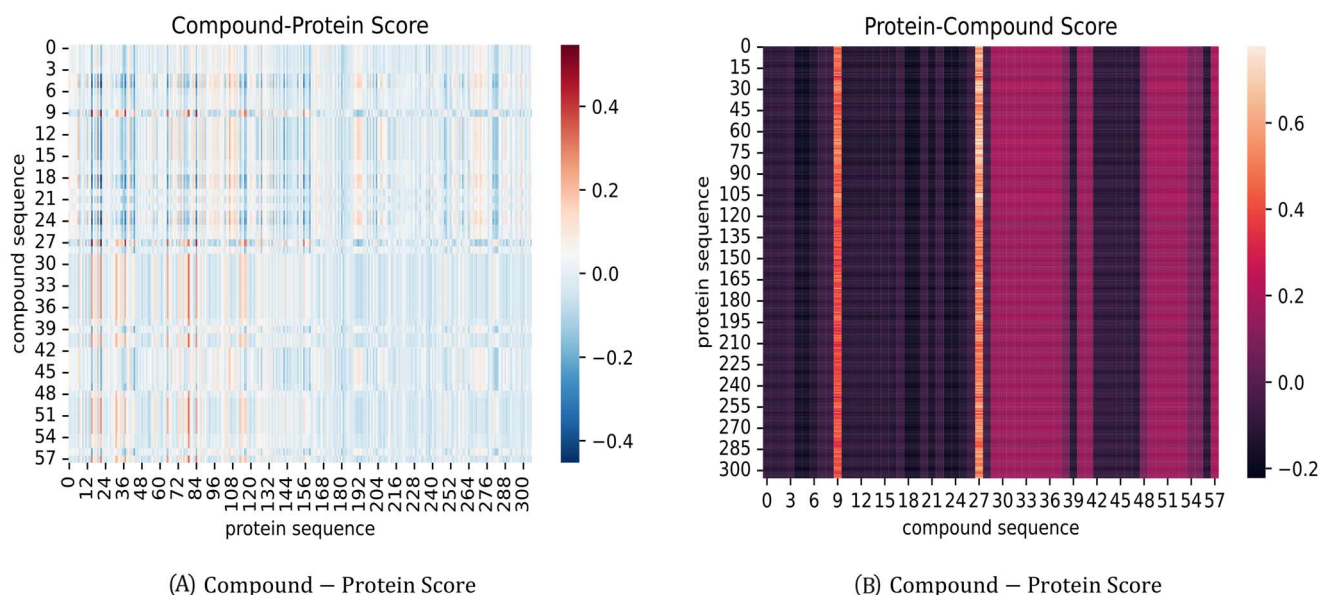


Figure 6. Visualization of attention matrices of GC373 (PDB ID: 6WTK).

enlightening observation is that the covalent bond is reversible, and the reversible reaction with thiols can effectively inhibit cysteine protease since they can bind effectively in the active site of the target protein. It indicates that the co-attention mechanism can elucidate meaningful interaction information. The detailed description of the co-crystal complex of 3CL M^{Pro} and ML188 (PDB ID: 7L0D) can be found in the supplementary.

Conclusion

In this study, we proposed an end-end deep learning framework CoaDTI to predict the DTI by fusing the multi-modal information. Since the protein input and drug input have different characteristics, CoaDTI incorporates the most suitable feature extractors to obtain the feature representations. For protein modality, transformer is employed to encode the input since it can capture the global information by focusing on different positions. For drug modality, GraphSage is one of the GNNs that can learn the structure information of the graph input, and aggregate the neighbourhood information, which can be applied to unseen graphs. In order to integrate the multi-modal information, we proposed co-attention modules instead of the concatenation process to leverage the cross-modality complementarity. We have designed three types of co-attention modules: stack, encoder-decoder and

intersection. In particular, the intersection co-attention module is interesting and intuitive because it can simulate the process of dynamic DTIs. A real-world issue is that the labelled DTI data are limited due to finance and labour restrictions despite the explosion of unlabelled data availability. The size of the dataset may dominate the performance of the model. We utilize the pre-trained transformer to obtain the external knowledge to address the issue. The experimental results show that CoaDTI can achieve the best performance on the three public datasets compared with SOTA models. The transfer learning strategy further boosts the performance to an unprecedented level. The performance of the CoaDTI-pro on the scarce dataset demonstrates that our proposed model has the capacity to identify the interaction of new targets or drugs with few labelled data. For future work, we plan to conduct more experiments to comprehensively evaluate CoaDTI and its variant CoaDTI-pro on other benchmark datasets such as KIBA [40] and Davis [41].

Our proposed framework also provides practical implications for downstream pharmaceutical tasks. CoaDTI can identify the candidate drugs for the specific target related to the disease such as COVID-19. Specifically, its annotation scores can reflect molecular binding mechanisms. The co-attention mechanism can enable CoaDTI to be interpretable. The attention scores can assist the researchers in pin-pointing the 3D binding site of the protein

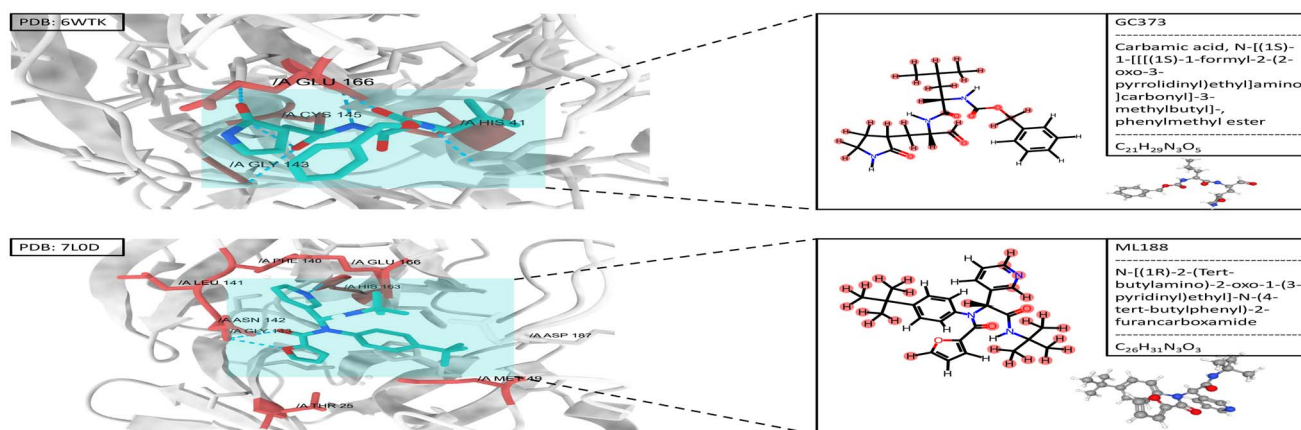


Figure 7. Visualization of the binding sites of 6WTK and 7L0D with highlighted residues and atoms. The left part is the 3D binding site attention visualization. The right part is the drug molecule structure attention visualization.

and the compound, and the identified key residues and atoms offer the fundamental cues for drug development.

Key Points

- An explainable deep learning framework is proposed to enable high-throughput drug–target binding annotation, which achieves the best performance among the SOTA models and provides insights for SARS-COV-2 applications.
- Our CoaDTI model treats the drug and protein as two different modes which can learn the information from the separate and shared space.
- The framework is based on pre-trained language models, boosting the performance to an unprecedented level with less training data.
- The co-attention mechanism in CoaDTI can measure the mutual effects of drugs and proteins. The researchers can gain insights by employing CoaDTI to elucidate where the model has focused on.

Supplementary information

Supplementary data are available at *Briefings in Bioinformatics* online.

Code availability

CoaDTI, CoaDTI-pro score code and their usage instructions are available at <https://github.com/Layne-Huang/CoaDTI>.

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